Example 3.40 An inductive coil has a resistance of 10 Ω and inductance of 100 mH. This coil is connected in parallel with a capacitor of 20 μF. A variable frequency power at 100 V is applied across this parallel circuit. Calculate the frequency at which the circuit will reasonate. Also calculate the Q-factor, dynamic impedance of the circuit, and resonant current.

Solution:

Resonant frequency, $$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

Substituting the given values,

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{1}{100\times10^{-3}\times20\times10^{-6}} - \frac{(10)^2}{(100\times10^{-3})^2}}$$

$$= 112.48\ \text{Hz}$$

$$\text{Q-factor} = \frac{2\pi f_0 L}{R} = \frac{6.28\times112.48\times100\times10^{-3}}{10} = 7.06$$

Dynamic impedance, $$Z_0 = \frac{L}{CR} = \frac{100\times10^{-3}}{20\times10^{-6}\times10} = 500\ \Omega$$

Current at resonance, $$I_0 = \frac{V}{Z_0} = \frac{100}{500} = 0.2\ \text{A}$$

Example 3.41 An inductive coil of resistance 5 Ω and inductive reactance 10 Ω is connected across a voltage of 230 V at 50 Hz. Calculate the value of the capacitor which when connected in parallel with the coil will bring down the magnitude of the circuit current to a minimum. Draw the phasor diagram.

Solution:

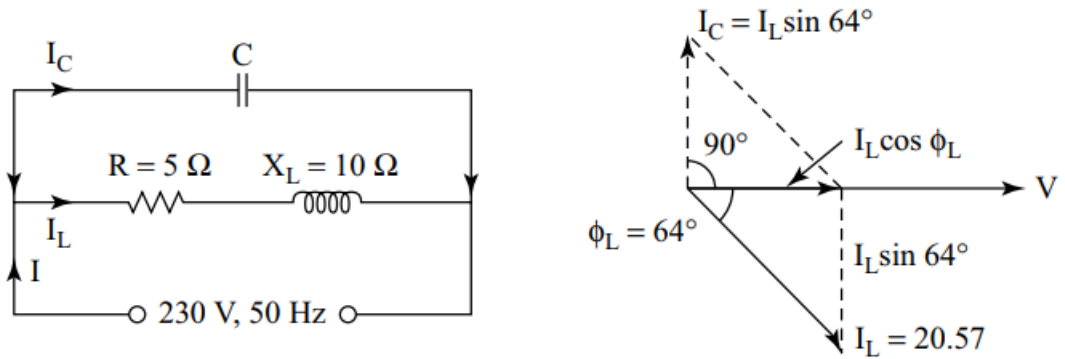


Figure 3.71

Before a capacitor is connected, current flowing through the inductor, I_L is

$$I_L = \frac{V}{Z} = \frac{230}{5 + j10} = \frac{230}{11.18 \angle 64°} = 20.57 \angle -64°$$

$$\cos \phi_L = \cos 64° = 0.438$$

$$\sin \phi_L = \sin 64° = 0.895$$

If a capacitor is now connected in parallel, it must draw a current I_C which will lead V by 90°. The magnitude of I_C must be equal to $I_L \sin \phi_L$ so that these two currents cancel each other. In such a case, the resultant current, I is the in-phase current, i.e., $I_L \cos \phi_L$.

$$I_C = I_L \sin \phi_L = 20.57 \times 0.895 = 18.4 \text{ A}$$

$$I_C = \frac{V}{X_C} \text{ or } X_C = \frac{V}{I_C} = \frac{230}{18.4} = 12.5\,\Omega$$

$$X_C = \frac{1}{\omega C} = 12.5$$

or,
$$C = \frac{1}{2\pi f \times 12.5} = \frac{1}{6.28 \times 50 \times 12.5} F = \frac{10^6}{314 \times 12.5} \mu F$$

$$= 254.7\,\mu F$$

Magnitude of the in-phase current, i.e., the current which is in phase with the voltage, $I = I_L \cos\phi$, is

$$I = I_L \cos\phi_L = 20.57 \times 0.438 = 9\,A$$

This is the minimum current drawn by the circuit and is called the resonant current, I_0.

Example 3.42 An inductor having a resistance of 4 Ω and an inductance of 20 mH are connected across a 230 V, 50 Hz supply. What value of capacitance should be connected in parallel to the inductor to produce a resonance condition? What will be the value of the resonant current?

Solution:

$$Z_L = R + j\omega L,\ \omega = 2\pi f = 2 \times 3.14 \times 50 = 314$$

$$Z_L = 4 + j314 \times 20 \times 10^{-3} = 4 + j6.28 = 7.44 \angle 57.5° \Omega$$

$$I_L = \frac{V}{Z_L} = \frac{230 \angle 0}{7.44 \angle 57.5°} = 30.9 \angle -57.5° \text{ A}$$

For resonance, the current drawn by the capacitor in parallel must be equal to $I_L \sin \phi_L$.

$$I_C = I_L \sin \phi_L = 30.9 \times \sin 57.5° = 30.9 \times 0.843 = 26 \text{ A}$$

$$I_C = \frac{V}{X_C} \text{ or, } X_C = \frac{V}{I_C} = \frac{230}{26} = 8.84 \Omega$$

$$X_C = \frac{1}{\omega C} = \frac{1}{314\,C} = 8.84 \Omega$$

or,

$$C = \frac{1}{314 \cdot 8.84} F = \frac{10^6}{314 \cdot 8.84} \mu F$$

$$= 360.2 \, \mu F$$

Resonant current for parallel resonance is the minimum current which is the in-phase component, i.e., $I_L \cos \phi_L$.

$$I_0 = I_L \cos \phi_L = 30.9 \times \cos 57.5° = 30.9 \times 0.537$$

$$= 16.6 \text{ A}$$

The phasor diagram representing the resonant condition has been shown in Fig. 3.72.

